NAME:_____

ID NUMBER: _____

**ECON 422**
**FINAL**

## Rules of the game

Count all the pages (you should have 12 pages including this) and write your name and ID number on the top of the page.

You have 2 and 1/5 hours to finish the test. The test is closed books and closed notes. You can use a calculator.

On the test you have the points assigned to each question. A correct answer with no explanation will NOT give you full credit. Make sure you explain each step. All relevant tables are at the end of the test.

I am very strict about cheating and I have a no tolerance policy. Your signature on the exam is your pledge to the honor code.

Q1. As you probably know there has been a lot of discussion about electricity demand in California. In this problem we are trying to explain the electricity consumption, as measured in kilowatt-hours per residential costumer in San Diego County. We are particularly interested in estimating the income and price elasticities of demand for electricity, studying whether there has been any structural change (break in the trend) and predicting the demand for 1993. The data are quarterly data from 1972:2 to 1992:4.

| | |
|---|---|
| KWH | = Electricity sales per residential costumers |
| PRICE | = Average price for the single-family rate tariff in real terms. (i.e. adjusted by inflation). |
| Y | = San Diego County total per-capita income in current terms |
| CDD | = Cooling degree days (average number of days in the quarter when the temperature was over 65) |
| HDD | = Heating degree days (average number of days in the quarter when the temperature was less than 65) |

Since we are interested in elasticities, I have transformed some of the data by taking the natural logarithm.

The basic model is (LKWH is the log of KWH, LY is the log of Y, and LPRICE is the log of PRICE)

Dependent Variable: LKWH
Method: Least Squares
Date: 05/06/02 Time: 17:39
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.435988 | 0.225173 | 1.936239 | 0.0565 |
| LY | -0.060167 | 0.158404 | -0.379830 | 0.7051 |
| LPRICE | -0.097068 | 0.029198 | -3.324449 | 0.0014 |
| CDD | 0.000262 | 3.53E-05 | 7.438805 | 0.0000 |
| HDD | 0.000360 | 3.09E-05 | 11.62252 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.663915 | Mean dependent var | 0.327688 |
| Adjusted R-squared | 0.646680 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.048270 | Akaike info criterion | -3.165648 |
| Sum squared resid | 0.181742 | Schwarz criterion | -3.019935 |
| Log likelihood | 136.3744 | F-statistic | 38.52102 |
| Durbin-Watson stat | 1.196531 | Prob(F-statistic) | 0.000000 |

(a) (5pts)What are the income and price elasticities of demand for electricity? Do the signs coincide with your economic intuition?

The graph of LKWH also suggests the presence of a liner trend. I then decided to estimate the previous equation including a trend. At the same time, during the period before 1979 the price of crude oil escalated dramatically and conservation became an important issue for reducing demand. Buildings were better insulated, more energy-efficient appliances and machinery were built and automobiles became more fuel-efficient. One may therefore expect that there is a break in trend in 1979. I construct a dummy variable (DBROKEN) that has zeros before 1979:1 and ones after that. TDBROKEN is an interactive dummy variable constructed as time*dbroken.

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.460403 | 0.202617 | 2.272285 | 0.0259 |
| DBROKEN | 0.158074 | 0.036955 | 4.277478 | 0.0001 |
| TREND | 0.004933 | 0.001145 | 4.309615 | 0.0000 |
| TDBROKEN | -0.006643 | 0.001280 | -5.188350 | 0.0000 |
| LY | 0.009295 | 0.149914 | 0.062002 | 0.9507 |
| LPRICE | -0.177869 | 0.034805 | -5.110525 | 0.0000 |
| CDD | 0.000260 | 2.93E-05 | 8.859899 | 0.0000 |
| HDD | 0.000357 | 2.57E-05 | 13.87170 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.776938 | Mean dependent var | 0.327688 |
| Adjusted Rsq | 0.756118 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.040104 | Akaike info criterion | -3.503272 |
| Sum squared resid | 0.120624 | Schwarz criterion | -3.270131 |
| Log likelihood | 153.3858 | F-statistic | 37.31839 |
| Durbin-Watson | 1.789111 | Prob(F-statistic) | 0.000000 |

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83

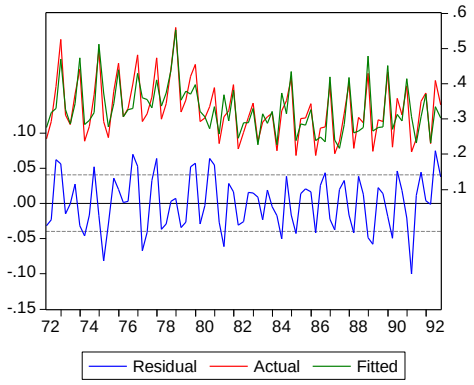
| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.280999 | 0.221801 | 1.266896 | 0.2090 |
| TREND | -0.000646 | 0.000223 | -2.893065 | 0.0050 |
| LY | 0.045459 | 0.155752 | 0.291869 | 0.7712 |
| LPRICE | -0.073426 | 0.029081 | -2.524867 | 0.0136 |
| CDD | 0.000262 | 3.37E-05 | 7.781580 | 0.0000 |
| HDD | 0.000357 | 2.96E-05 | 12.07104 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.696865 | Mean dependent var | 0.327688 |
| Adjusted Rsq | 0.677181 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.046140 | Akaike info criterion | -3.244739 |
| Sum squared resid | 0.163924 | Schwarz criterion | -3.069883 |
| Log likelihood | 140.6567 | F-statistic | 35.40251 |
| Durbin-Watson | 1.323347 | Prob(F-statistic) | 0.000000 |

(b) (10pts) Is there a significance difference in the trend before and after 1979? Do you find evidence of a broken trend?

(c) (10pts) Explain in economic terms the difference between the trend before and after 1979. Draw a graph showing the shape of the trend suggested by the regression above.

(d) (10 pts) The graph of the residuals from the above regression suggests that we have not taken into account seasonality. I then created seasonal dummy variables and I included them in the regression. Is seasonality significant? Comment on the results from this new regression.




Dependent Variable: LKWH
Method: Least Squares
Date: 05/06/02 Time: 18:03
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.569909 | 0.132071 | 4.315179 | 0.0000 |
| DBROKEN | 0.144370 | 0.024410 | 5.914400 | 0.0000 |
| TREND | 0.004153 | 0.000759 | 5.471034 | 0.0000 |
| TDBROKEN | -0.005892 | 0.000849 | -6.939062 | 0.0000 |
| LY | 0.014437 | 0.097710 | 0.147750 | 0.8830 |
| LPRICE | -0.182071 | 0.022515 | -8.086597 | 0.0000 |
| CDD | 0.000208 | 3.29E-05 | 6.311122 | 0.0000 |
| HDD | 0.000194 | 3.36E-05 | 5.773004 | 0.0000 |
| D1 | 0.035064 | 0.017212 | 2.037193 | 0.0453 |
| D2 | -0.077713 | 0.008723 | -8.909389 | 0.0000 |
| D3 | -0.064836 | 0.017090 | -3.793822 | 0.0003 |

| | | | |
|---|---|---|---|
| R-squared | 0.910562 | Mean dependent var | 0.327688 |
| Adjusted R-squared | 0.898140 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.025918 | Akaike info criterion | -4.344890 |
| Sum squared resid | 0.048365 | Schwarz criterion | -4.024321 |
| Log likelihood | 191.3130 | F-statistic | 73.30283 |
| Durbin-Watson stat | 1.782561 | Prob(F-statistic) | 0.000000 |

(e) (10pts)Using the appropriate regression from the ones provided below, test whether the second and third quarters have the same effect on demand for electricity. (note d23=d2+d3, and d12=d1+d2)

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.522760 | 0.167239 | 3.125834 | 0.0025 |
| DBROKEN | 0.170321 | 0.030565 | 5.572372 | 0.0000 |
| TREND | 0.005056 | 0.000948 | 5.335962 | 0.0000 |
| TDBROKEN | -0.006890 | 0.001060 | -6.499535 | 0.0000 |
| LY | -0.016886 | 0.123762 | -0.136440 | 0.8918 |
| LPRICE | -0.176381 | 0.028531 | -6.182202 | 0.0000 |
| CDD | 0.000195 | 4.17E-05 | 4.686642 | 0.0000 |
| HDD | 0.000389 | 2.17E-05 | 17.88203 | 0.0000 |
| D12 | -0.065624 | 0.010824 | -6.062740 | 0.0000 |
| D3 | -0.003958 | 0.018393 | -0.215165 | 0.8302 |

| | | | |
|---|---|---|---|
| R-squared | 0.854188 | Mean dependent var | 0.327688 |
| Adjusted Rsq | 0.836211 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.032865 | Akaike info criterion | -3.880213 |
| Sum squared resid | 0.078850 | Schwarz criterion | -3.588786 |
| Log likelihood | 171.0288 | F-statistic | 47.51606 |
| Durbin-Watson | 1.607283 | Prob(F-statistic) | 0.000000 |

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.562031 | 0.131016 | 4.289776 | 0.0001 |
| DBROKEN | 0.141441 | 0.023908 | 5.916086 | 0.0000 |
| TREND | 0.004059 | 0.000743 | 5.466030 | 0.0000 |
| TDBROKEN | -0.005784 | 0.000830 | -6.969598 | 0.0000 |
| LY | 0.022691 | 0.096526 | 0.235071 | 0.8148 |
| LPRICE | -0.182601 | 0.022413 | -8.146981 | 0.0000 |
| CDD | 0.000225 | 2.09E-05 | 10.76393 | 0.0000 |
| HDD | 0.000185 | 3.08E-05 | 6.016854 | 0.0000 |
| D1 | 0.041949 | 0.013619 | 3.080221 | 0.0029 |
| D23 | -0.074910 | 0.007584 | -9.877653 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.910024 | Mean dependent var | 0.327688 |
| Adjusted Rsq | 0.898931 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.025817 | Akaike info criterion | -4.362983 |
| Sum squared resid | 0.048656 | Schwarz criterion | -4.071557 |
| Log likelihood | 191.0638 | F-statistic | 82.03599 |
| Durbin-Watson | 1.820586 | Prob(F-statistic) | 0.000000 |

(f) (10 pts) Suppose now that we had started our problem by modeling seasonality first. Attached is the correlogram from the residuals of the model with only seasonal dummies. Only looking at the correlogram, which model for the cycles would you consider (AR, MA, ARMA)? Why?

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.343682 | 0.010582 | 32.47851 | 0.0000 |
| D1 | 0.081656 | 0.015151 | 5.389496 | 0.0000 |
| D2 | -0.099017 | 0.014965 | -6.616558 | 0.0000 |
| D3 | -0.041967 | 0.014965 | -2.804346 | 0.0063 |

| | | | |
|---|---|---|---|
| R-squared | 0.656472 | Mean dependent var | 0.327688 |
| Adjusted R-squared | 0.643427 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.048492 | Akaike info criterion | -3.167841 |
| Sum squared resid | 0.185767 | Schwarz criterion | -3.051270 |
| Log likelihood | 135.4654 | F-statistic | 50.32225 |
| Durbin-Watson stat | 0.893338 | Prob(F-statistic) | 0.000000 |

Date: 05/07/02 Time: 00:35
Sample: 1972:2 1992:4
Included observations: 83

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.553 | 0.553 | 26.336 | 0.000 |
| | | 2 | 0.435 | 0.185 | 42.793 | 0.000 |
| | | 3 | 0.417 | 0.177 | 58.153 | 0.000 |
| | | 4 | 0.459 | 0.217 | 77.000 | 0.000 |
| | | 5 | 0.364 | -0.003 | 89.013 | 0.000 |
| | | 6 | 0.281 | -0.040 | 96.271 | 0.000 |
| | | 7 | 0.304 | 0.076 | 104.88 | 0.000 |
| | | 8 | 0.438 | 0.255 | 122.93 | 0.000 |
| | | 9 | 0.291 | -0.116 | 131.02 | 0.000 |
| | | 10 | 0.193 | -0.099 | 134.61 | 0.000 |
| | | 11 | 0.194 | -0.013 | 138.29 | 0.000 |
| | | 12 | 0.258 | 0.053 | 144.91 | 0.000 |
| | | 13 | 0.188 | -0.032 | 148.46 | 0.000 |
| | | 14 | 0.182 | 0.091 | 151.85 | 0.000 |
| | | 15 | 0.098 | -0.139 | 152.84 | 0.000 |
| | | 16 | 0.154 | -0.027 | 155.34 | 0.000 |
| | | 17 | 0.044 | -0.120 | 155.55 | 0.000 |
| | | 18 | 0.048 | 0.069 | 155.80 | 0.000 |
| | | 19 | 0.037 | 0.006 | 155.95 | 0.000 |
| | | 20 | 0.013 | -0.096 | 155.97 | 0.000 |
| | | 21 | 0.036 | 0.073 | 156.11 | 0.000 |
| | | 22 | 0.056 | 0.033 | 156.47 | 0.000 |
| | | 23 | 0.050 | 0.065 | 156.77 | 0.000 |
| | | 24 | 0.014 | -0.056 | 156.79 | 0.000 |

(g) (10pts) I attached few regressions that I estimated for different possible models for the cycles of this data. Of the models that I estimated which one do you think is the best model and why?

(h) (20pts) Given your answer in (g) compute your forecast for the first and second quarter of 1993. Below are some of the data you may need to compute the forecasts depending on which model you have choosen in the previous question?

| Date | LKWH | Residuals from MA(1) model | Residuals from MA(2) model | Residuals from ARMA(1,1) model |
|---|---|---|---|---|
| 1991:1 | 0.392801 | -0.005694076 | -0.022133062 | -0.026877354 |
| 1991:2 | 0.205318 | -0.036940399 | -0.012792849 | -0.025993192 |
| 1991:3 | 0.242447 | -0.043961233 | -0.047096968 | -0.038930947 |
| 1991:4 | 0.349169 | 0.023702385 | 0.031645883 | 0.035213591 |
| 1992:1 | 0.373517 | -0.060759518 | -0.054551938 | -0.034587082 |
| 1992:2 | 0.229931 | 0.010488942 | 0.004106577 | 0.011495529 |
| 1992:3 | 0.408811 | 0.102749556 | 0.119780918 | 0.128869546 |
| 1992:4 | 0.338866 | -0.04739108 | -0.064279571 | -0.020704755 |

Q3 (5 pts) Compute $VAR(y_t)$ where $y_t = \varepsilon_t + 0.8\varepsilon_{t-1} + 0.2\varepsilon_{t-2}$. Assume $\varepsilon_t \sim W.N.$

Q3. (10 pts) Compute $\gamma(1)$, $\gamma(2)$, $\rho(1)$, $\rho(2)$ for $y_t = \varepsilon_t + 0.8\varepsilon_{t-1}$. Assume $\varepsilon_t \sim W.N.$

BONUS QUESTION

(5pts)Show that a W.N. process with mean 0 and variance $\sigma^2$ is a stationary process.

Dependent Variable: LKWH
Method: Least Squares
Sample(adjusted): 1972:3 1992:4
Included observations: 82 after adjusting endpoints
Convergence achieved after 3 iterations

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.343680 | 0.011801 | 29.12412 | 0.0000 |
| D1 | 0.081524 | 0.009874 | 8.256127 | 0.0000 |
| D2 | -0.099022 | 0.011185 | -8.853110 | 0.0000 |
| D3 | -0.041969 | 0.009719 | -4.318442 | 0.0000 |
| AR(1) | 0.553337 | 0.094933 | 5.828729 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.758427 | Mean dependent var | 0.328716 |
| Adjusted R-squared | 0.745877 | S.D. dependent var | 0.081161 |
| S.E. of regression | 0.040914 | Akaike info criterion | -3.495655 |
| Sum squared resid | 0.128894 | Schwarz criterion | -3.348904 |
| Log likelihood | 148.3219 | F-statistic | 60.43594 |
| Durbin-Watson stat | 2.170212 | Prob(F-statistic) | 0.000000 |

| | | |
|---|---|---|
| Inverted AR Roots | .55 | |

Dependent Variable: LKWH
Method: Least Squares
Sample(adjusted): 1972:4 1992:4
Included observations: 81 after adjusting endpoints
Convergence achieved after 3 iterations

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.344325 | 0.013965 | 24.65560 | 0.0000 |
| D1 | 0.082498 | 0.010160 | 8.120004 | 0.0000 |
| D2 | -0.098858 | 0.009933 | -9.951948 | 0.0000 |
| D3 | -0.041520 | 0.010060 | -4.127323 | 0.0001 |
| AR(1) | 0.440392 | 0.115352 | 3.817796 | 0.0003 |
| AR(2) | 0.202929 | 0.119283 | 1.701235 | 0.0930 |

| | | | |
|---|---|---|---|
| R-squared | 0.766959 | Mean dependent var | 0.329164 |
| Adjusted R-squared | 0.751423 | S.D. dependent var | 0.081565 |
| S.E. of regression | 0.040666 | Akaike info criterion | -3.495649 |
| Sum squared resid | 0.124031 | Schwarz criterion | -3.318282 |
| Log likelihood | 147.5738 | F-statistic | 49.36631 |
| Durbin-Watson stat | 2.019048 | Prob(F-statistic) | 0.000000 |

| | | |
|---|---|---|
| Inverted AR Roots | .72 | -.28 |

Dependent Variable: LKWH
Method: Least Squares
Included observations: 80 after adjusting endpoints
Convergence achieved after 3 iterations

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.341038 | 0.016705 | 20.41491 | 0.0000 |
| D1 | 0.083606 | 0.009972 | 8.384445 | 0.0000 |
| D2 | -0.095857 | 0.011303 | -8.480916 | 0.0000 |
| D3 | -0.039295 | 0.009949 | -3.949690 | 0.0002 |
| AR(1) | 0.415956 | 0.115128 | 3.612978 | 0.0006 |
| AR(2) | 0.101962 | 0.133497 | 0.763781 | 0.4475 |
| AR(3) | 0.191797 | 0.122323 | 1.567958 | 0.1212 |

| | | | |
|---|---|---|---|
| R-squared | 0.778065 | Mean dependent var | 0.328389 |
| Adjusted R-squared | 0.759823 | S.D. dependent var | 0.081779 |
| S.E. of regression | 0.040078 | Akaike info criterion | -3.512548 |
| Sum squared resid | 0.117256 | Schwarz criterion | -3.304121 |
| Log likelihood | 147.5019 | F-statistic | 42.65412 |
| Durbin-Watson stat | 2.088123 | Prob(F-statistic) | 0.000000 |

| | | | |
|---|---|---|---|
| Inverted AR Roots | .82 | -.20+.44i | -.20 -.44i |

Dependent Variable: LKWH
Sample: 1972:2 1992:4
Included observations: 83
Convergence achieved after 16 iterations
Backcast: 1972:1

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.343682 | 0.010208 | 33.66780 | 0.0000 |
| D1 | 0.080773 | 0.011778 | 6.857889 | 0.0000 |
| D2 | -0.099064 | 0.014462 | -6.850066 | 0.0000 |
| D3 | -0.041967 | 0.011606 | -3.615951 | 0.0005 |
| MA(1) | 0.414360 | 0.105679 | 3.920922 | 0.0002 |

| | | | |
|---|---|---|---|
| R-squared | 0.730621 | Mean dependent var | 0.327688 |
| Adjusted R-squared | 0.716806 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.043215 | Akaike info criterion | -3.386892 |
| Sum squared resid | 0.145670 | Schwarz criterion | -3.241179 |
| Log likelihood | 145.5560 | F-statistic | 52.88864 |
| Durbin-Watson stat | 1.727906 | Prob(F-statistic) | 0.000000 |

| | |
|---|---|
| Inverted MA Roots | -.41 |

Dependent Variable: LKWH
Method: Least Squares
Sample: 1972:2 1992:4
Included observations: 83
Convergence achieved after 6 iterations
Backcast: 1971:4 1972:1

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.343847 | 0.010432 | 32.96238 | 0.0000 |
| D1 | 0.081522 | 0.010949 | 7.445304 | 0.0000 |
| D2 | -0.100036 | 0.011460 | -8.729270 | 0.0000 |
| D3 | -0.042131 | 0.010729 | -3.926839 | 0.0002 |
| MA(1) | 0.485835 | 0.113232 | 4.290606 | 0.0001 |
| MA(2) | 0.269121 | 0.120351 | 2.236143 | 0.0282 |

| | | | |
|---|---|---|---|
| R-squared | 0.750871 | Mean dependent var | 0.327688 |
| Adjusted R-squared | 0.734694 | S.D. dependent var | 0.081208 |
| S.E. of regression | 0.041828 | Akaike info criterion | -3.440946 |
| Sum squared resid | 0.134719 | Schwarz criterion | -3.266090 |
| Log likelihood | 148.7992 | F-statistic | 46.41538 |
| Durbin-Watson stat | 1.924393 | Prob(F-statistic) | 0.000000 |

| | | |
|---|---|---|
| Inverted MA Roots | -.24+.46i | -.24 -.46i |

Dependent Variable: LKWH
Method: Least Squares
Date: 05/06/02 Time: 18:16
Sample(adjusted): 1972:3 1992:4
Included observations: 82 after adjusting endpoints
Convergence achieved after 10 iterations
Backcast: 1972:2

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.340657 | 0.022283 | 15.28752 | 0.0000 |
| D1 | 0.082325 | 0.010617 | 7.754019 | 0.0000 |
| D2 | -0.098766 | 0.010655 | -9.269144 | 0.0000 |
| D3 | -0.042208 | 0.010468 | -4.032260 | 0.0001 |
| AR(1) | 0.926149 | 0.059771 | 15.49485 | 0.0000 |
| MA(1) | -0.646372 | 0.122486 | -5.277088 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.780985 | Mean dependent var | 0.328716 |
| Adjusted R-squared | 0.766576 | S.D. dependent var | 0.081161 |
| S.E. of regression | 0.039212 | Akaike info criterion | -3.569297 |
| Sum squared resid | 0.116858 | Schwarz criterion | -3.393196 |
| Log likelihood | 152.3412 | F-statistic | 54.20159 |
| Durbin-Watson stat | 1.855379 | Prob(F-statistic) | 0.000000 |

| | |
|---|---|
| Inverted AR Roots | .93 |
| Inverted MA Roots | .65 |

## Upper 5% Points of the *F*-Distribution

| $n$ \ $m$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 12 | 15 | 20 | 24 | 30 | 40 | 60 | 120 | ∞ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 | 241.9 | 243.9 | 245.9 | 248.0 | 249.1 | 250.1 | 251.1 | 252.2 | 253.3 | 254.3 |
| 2 | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 | 19.40 | 19.41 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50 |
| 3 | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 | 8.79 | 8.74 | 8.70 | 8.66 | 8.64 | 8.62 | 8.59 | 8.57 | 8.55 | 8.53 |
| 4 | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 | 5.96 | 5.91 | 5.86 | 5.80 | 5.77 | 5.75 | 5.72 | 5.69 | 5.66 | 5.63 |
| 5 | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 | 4.74 | 4.68 | 4.62 | 4.56 | 4.53 | 4.50 | 4.46 | 4.43 | 4.40 | 4.36 |
| 6 | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 | 4.06 | 4.00 | 3.94 | 3.87 | 3.84 | 3.81 | 3.77 | 3.74 | 3.70 | 3.67 |
| 7 | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 | 3.64 | 3.57 | 3.51 | 3.44 | 3.41 | 3.38 | 3.34 | 3.30 | 3.27 | 3.23 |
| 8 | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 | 3.35 | 3.28 | 3.22 | 3.15 | 3.12 | 3.08 | 3.04 | 3.01 | 2.97 | 2.93 |
| 9 | 5.12 | 4.26 | 3.86 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 | 3.14 | 3.07 | 3.01 | 2.94 | 2.90 | 2.86 | 2.83 | 2.79 | 2.75 | 2.71 |
| 10 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 | 2.98 | 2.91 | 2.85 | 2.77 | 2.74 | 2.70 | 2.66 | 2.62 | 2.58 | 2.54 |
| 11 | 4.84 | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2.95 | 2.90 | 2.85 | 2.79 | 2.72 | 2.65 | 2.61 | 2.57 | 2.53 | 2.49 | 2.45 | 2.40 |
| 12 | 4.75 | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2.85 | 2.80 | 2.75 | 2.69 | 2.62 | 2.54 | 2.51 | 2.47 | 2.43 | 2.38 | 2.34 | 2.30 |
| 13 | 4.67 | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 | 2.67 | 2.60 | 2.53 | 2.46 | 2.42 | 2.38 | 2.34 | 2.30 | 2.25 | 2.21 |
| 14 | 4.60 | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 | 2.60 | 2.53 | 2.46 | 2.39 | 2.35 | 2.31 | 2.27 | 2.22 | 2.18 | 2.13 |
| 15 | 4.54 | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 | 2.48 | 2.40 | 2.33 | 2.29 | 2.25 | 2.20 | 2.16 | 2.11 | 2.07 |
| 16 | 4.49 | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 | 2.42 | 2.35 | 2.28 | 2.24 | 2.19 | 2.15 | 2.11 | 2.06 | 2.01 |
| 17 | 4.45 | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 | 2.38 | 2.31 | 2.23 | 2.19 | 2.15 | 2.10 | 2.06 | 2.01 | 1.96 |
| 18 | 4.41 | 3.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 | 2.34 | 2.27 | 2.19 | 2.15 | 2.11 | 2.06 | 2.02 | 1.97 | 1.92 |
| 19 | 4.38 | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 | 2.31 | 2.23 | 2.16 | 2.11 | 2.07 | 2.03 | 1.98 | 1.93 | 1.88 |
| 20 | 4.35 | 3.49 | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 | 2.35 | 2.28 | 2.20 | 2.12 | 2.08 | 2.04 | 1.99 | 1.95 | 1.90 | 1.84 |
| 21 | 4.32 | 3.47 | 3.07 | 2.84 | 2.68 | 2.57 | 2.49 | 2.42 | 2.37 | 2.32 | 2.25 | 2.18 | 2.10 | 2.05 | 2.01 | 1.96 | 1.92 | 1.87 | 1.81 |
| 22 | 4.30 | 3.44 | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 | 2.30 | 2.23 | 2.15 | 2.07 | 2.03 | 1.98 | 1.94 | 1.89 | 1.84 | 1.78 |
| 23 | 4.28 | 3.42 | 3.03 | 2.80 | 2.64 | 2.53 | 2.44 | 2.37 | 2.32 | 2.27 | 2.20 | 2.13 | 2.05 | 2.01 | 1.96 | 1.91 | 1.86 | 1.81 | 1.76 |
| 24 | 4.26 | 3.40 | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 | 2.25 | 2.18 | 2.11 | 2.03 | 1.98 | 1.94 | 1.89 | 1.84 | 1.79 | 1.73 |
| 25 | 4.24 | 3.39 | 2.99 | 2.76 | 2.60 | 2.49 | 2.40 | 2.34 | 2.28 | 2.24 | 2.16 | 2.09 | 2.01 | 1.96 | 1.92 | 1.87 | 1.82 | 1.77 | 1.71 |
| 26 | 4.23 | 3.37 | 2.98 | 2.74 | 2.59 | 2.47 | 2.39 | 2.32 | 2.27 | 2.22 | 2.15 | 2.07 | 1.99 | 1.95 | 1.90 | 1.85 | 1.80 | 1.75 | 1.69 |
| 27 | 4.21 | 3.35 | 2.96 | 2.73 | 2.57 | 2.46 | 2.37 | 2.31 | 2.25 | 2.20 | 2.13 | 2.06 | 1.97 | 1.93 | 1.88 | 1.84 | 1.79 | 1.73 | 1.67 |
| 28 | 4.20 | 3.34 | 2.95 | 2.71 | 2.56 | 2.45 | 2.36 | 2.29 | 2.24 | 2.19 | 2.12 | 2.04 | 1.96 | 1.91 | 1.87 | 1.82 | 1.77 | 1.71 | 1.65 |
| 29 | 4.18 | 3.33 | 2.93 | 2.70 | 2.55 | 2.43 | 2.35 | 2.28 | 2.22 | 2.18 | 2.10 | 2.03 | 1.94 | 1.90 | 1.85 | 1.81 | 1.75 | 1.70 | 1.64 |
| 30 | 4.17 | 3.32 | 2.92 | 2.69 | 2.53 | 2.42 | 2.33 | 2.27 | 2.21 | 2.16 | 2.09 | 2.01 | 1.93 | 1.89 | 1.84 | 1.79 | 1.74 | 1.68 | 1.62 |
| 40 | 4.08 | 3.23 | 2.84 | 2.61 | 2.45 | 2.34 | 2.25 | 2.18 | 2.12 | 2.08 | 2.00 | 1.92 | 1.84 | 1.79 | 1.74 | 1.69 | 1.64 | 1.58 | 1.51 |
| 60 | 4.00 | 3.15 | 2.76 | 2.53 | 2.37 | 2.25 | 2.17 | 2.10 | 2.04 | 1.99 | 1.92 | 1.84 | 1.75 | 1.70 | 1.65 | 1.59 | 1.53 | 1.47 | 1.39 |
| 120 | 3.92 | 3.07 | 2.68 | 2.45 | 2.29 | 2.17 | 2.09 | 2.02 | 1.96 | 1.91 | 1.83 | 1.75 | 1.66 | 1.61 | 1.55 | 1.50 | 1.43 | 1.35 | 1.25 |
| ∞ | 3.84 | 3.00 | 2.60 | 2.37 | 2.21 | 2.10 | 2.01 | 1.94 | 1.88 | 1.83 | 1.75 | 1.67 | 1.57 | 1.52 | 1.46 | 1.39 | 1.32 | 1.22 | 1.00 |

*Note:* $m$ = degrees of freedom for the numerator
$n$ = degrees of freedom for the denominator

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**Percentage Points of the *t*-Distribution**

| d.f. | 1T = 0.4<br>2T = 0.8 | 0.25<br>0.5 | 0.1<br>0.2 | 0.05<br>0.1 | 0.025<br>0.05 | 0.01<br>0.02 | 0.005<br>0.01 | 0.0025<br>0.005 | 0.001<br>0.002 | 0.0005<br>0.001 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.325 | 1.000 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 127.32 | 318.31 | 636.62 |
| 2 | .289 | 0.816 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 14.089 | 22.327 | 31.598 |
| 3 | .277 | .765 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 7.453 | 10.214 | 12.924 |
| 4 | .271 | .741 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 5.598 | 7.173 | 8.610 |
| 5 | 0.267 | 0.727 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 4.773 | 5.893 | 6.869 |
| 6 | .265 | .718 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 4.317 | 5.208 | 5.959 |
| 7 | .263 | .711 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.029 | 4.785 | 5.408 |
| 8 | .262 | .706 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 3.833 | 4.501 | 5.041 |
| 9 | .261 | .703 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 3.690 | 4.297 | 4.781 |
| 10 | 0.260 | 0.700 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 3.581 | 4.144 | 4.587 |
| 11 | .260 | .697 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 3.497 | 4.025 | 4.437 |
| 12 | .259 | .695 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.428 | 3.930 | 4.318 |
| 13 | .259 | .694 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.372 | 3.852 | 4.221 |
| 14 | .258 | .692 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.326 | 3.787 | 4.140 |
| 15 | 0.258 | 0.691 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.286 | 3.733 | 4.073 |
| 16 | .258 | .690 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.252 | 3.686 | 4.015 |
| 17 | .257 | .689 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.222 | 3.646 | 3.965 |
| 18 | .257 | .688 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.197 | 3.610 | 3.922 |
| 19 | .257 | .688 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.174 | 3.579 | 3.883 |
| 20 | 0.257 | 0.687 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.153 | 3.552 | 3.850 |
| 21 | .257 | .686 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.135 | 3.527 | 3.819 |
| 22 | .256 | .686 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.119 | 3.505 | 3.792 |
| 23 | .256 | .685 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.104 | 3.485 | 3.767 |
| 24 | .256 | .685 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.091 | 3.467 | 3.745 |
| 25 | 0.256 | 0.684 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.078 | 3.450 | 3.725 |
| 26 | .256 | .684 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.067 | 3.435 | 3.707 |
| 27 | .256 | .684 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.057 | 3.421 | 3.690 |
| 28 | .256 | .683 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.047 | 3.408 | 3.674 |
| 29 | .256 | .683 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.038 | 3.396 | 3.659 |
| 30 | 0.256 | 0.683 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.030 | 3.385 | 3.646 |
| 40 | .255 | .681 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 2.971 | 3.307 | 3.551 |
| 60 | .254 | .679 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 2.915 | 3.232 | 3.460 |
| 120 | .254 | .677 | 1.289 | 1.658 | 1.980 | 2.358 | 2.617 | 2.860 | 3.160 | 3.373 |
| ∞ | .253 | .674 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 2.807 | 3.090 | 3.291 |

*Note:* 1T = area under one tail; 2T = area under both tails.

For 25 degrees of freedom (d.f.), $P(t > 2.060) = 0.025$ and $P(t < -2.060 \text{ or } t > 2.060) = 0.05$.

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